

LAB # 2

Networking Basics Commands

1. PC Network TCP/IP Configuration

Objectives

- Gather information including connection, host name, Layer 2 MAC address and Layer 3 TCP/IP network address information.
- Compare network information to other PCs on the network.
- Learn to use the TCP/IP Packet Internet Groper (**ping**) command from a workstation.
- Learn to use the Trace Route (**tracert**) command from a workstation.

Background

This lab assumes the use of any version of Windows. This is a non-destructive lab and can be done on any machine without concern of changing the system configuration. Ideally, this lab is performed in a Computer Lab or other LAN environment that connects to the Internet.

Step 1: Connect into the Internet

Establish and verify connectivity to the Internet. This ensures the computer has an IP address.

Step 2: Gather TCP/IP configuration information

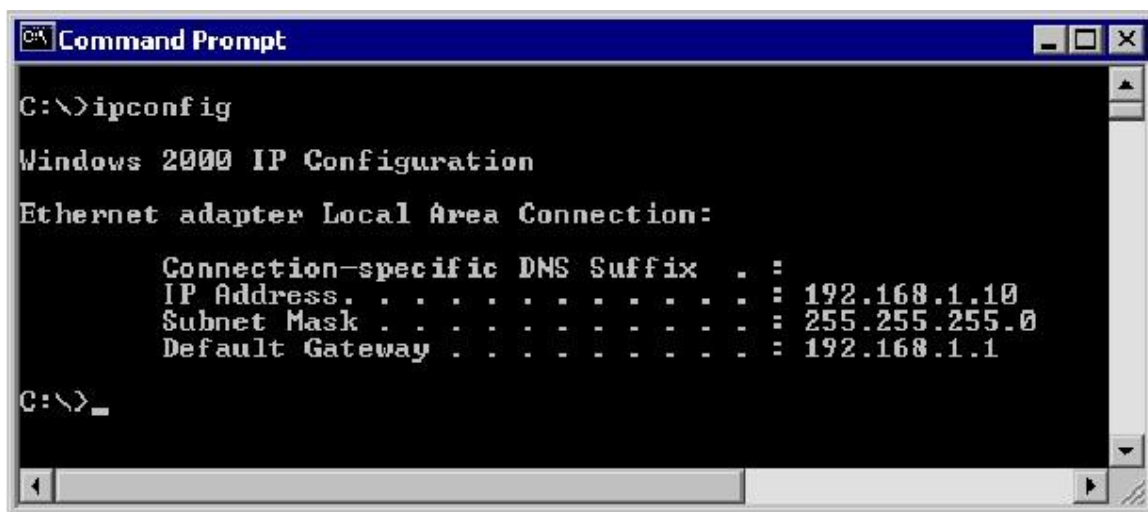
Use the Start menu to open Command Prompt, an MS-DOS-like window.

Press **Start > Command Prompt**

OR

Press **Start > Run Then type cmd.**

The following figure shows the Command screen. Type **ipconfig** and press the **Enter** key. The spelling of **ipconfig** is critical while case is not. It is short term for IP Configuration.



```
C:\>ipconfig

Windows 2000 IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . .               : 192.168.1.10
    Subnet Mask . . . . .             : 255.255.255.0
    Default Gateway . . . . .         : 192.168.1.1

C:\>_
```

Fig 2.1 ipconfig on Command Prompt

This first screen shows the IP address, subnet mask, and default gateway. The IP address and the default gateway should be in the same network or subnet, otherwise this host would not be able to communicate outside the network. In the figure the subnet mask tells us that the first three octets must be the same to be in the same network.

Step 3: Record the following TCP/IP information for this computer

IP address: _____

Subnet Mask: _____

Default Gateway: _____

Step 4: Compare the TCP/IP configuration of this computer to others on the LAN

If this computer is on a LAN, compare the information of several machines.

Are there any similarities? _____

What is similar about the IP addresses? _____

What is similar about the default gateways? _____

The IP addresses should share the same network portion. All machines in the LAN should share the same default gateway.

Record a couple of the IP Addresses:

Step 5: Check additional TCP/IP configuration information

To see detailed information, type **ipconfig /all** and press Enter. The figure shows the detailed IP configuration screen.

```
C:\>ipconfig /all

Windows 2000 IP Configuration

    Host Name . . . . . : thunder
    Primary DNS Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . :
    Description . . . . . : LNE100TX Fast Ethernet Adapter Version 1.0
    Physical Address. . . . . : 00-A0-CC-23-FE-40
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IP Address. . . . . : 192.168.1.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
    DHCP Server . . . . . : 192.168.1.1
    DNS Servers . . . . . : 24.0.224.33
                           24.0.224.34
    Lease Obtained. . . . . : Tuesday, January 09, 2001 10:56:19 AM
    Lease Expires . . . . . : Monday, January 15, 2001 10:56:19 AM

C:\>
```

Fig 2.2 ipconfig /all on Command Prompt

The host name, including the computer name and NetBIOS name should be displayed. Also, the DHCP server address, if used, and the date the IP lease starts and ends should be displayed. Look over the information. Entries for the DNS, used in name resolution servers, may also be present.

The previous figure reveals that the router is performing both DHCP and DNS services for this network. This would likely be a small office or home office (SOHO) or small branch office implementation.

Notice the Physical Address (MAC) and the NIC model (Description).

In the LAN, what similarities about the Physical (MAC) Addresses are seen?

While not a requirement, most LAN administrators try to standardize components like NICs.

Therefore, it would not be surprising to find all machines share the first three Hex pairs in the adapter address. These three pairs identify the manufacturer of the adapter.

Write down the IP addresses of any servers listed:

Write down the computer Host Name:

Write down the Host Names of a couple other computers:

Step 6: Close the screen

Close the screen when finished examining network settings.

Repeat the previous steps as necessary. Make sure that it is possible to return to and interpret this screen.

2. Using “ping” and “tracert”

Step 1: Establish and verify connectivity to the Internet

This ensures the computer has an IP address.

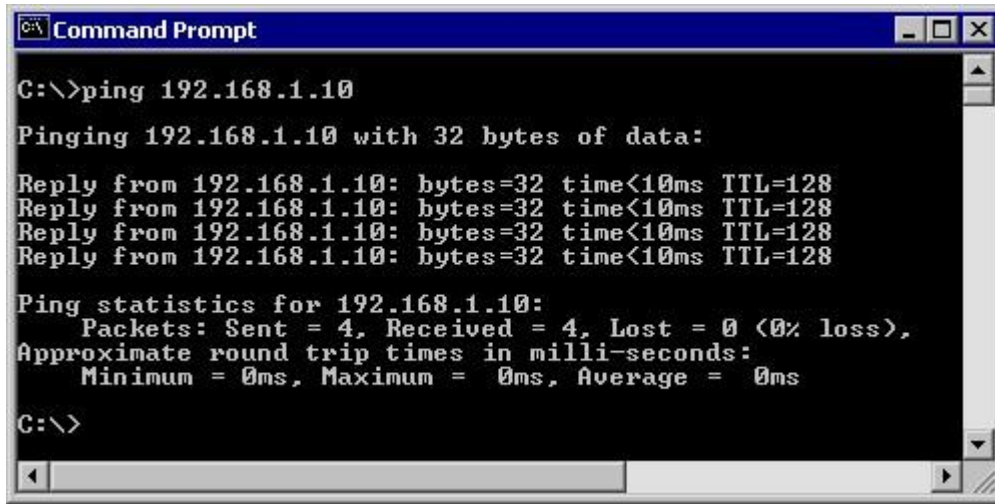
Step 2: Access the command prompt

As accessed in previous part of the lab.

Step 3: ping the IP address of another computer

In the window, type **ping**, a space, and the IP address of a computer recorded in the previous lab.

The following figure shows the successful results of **ping** to this IP address.



```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<10ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig 2.3 ping by Ip address on Command Prompt

ping uses the ICMP echo reply feature to test physical connectivity. Since **ping** reports on four attempts, it gives an indication of the reliability of the connection. Look over the results and verify that the **ping** was successful. Is the **ping** successful?

Step 4: ping the IP address of the default gateway

Try to **ping** the IP address of the default gateway if one was listed in the last exercise. If the **ping** is successful, it means there is physical connectivity to the router on the local network and probably the rest of the world.

Step 5: ping the IP address of a DHCP or DNS servers

Try to ping the IP address of any DHCP and/or DNS servers listed in the last exercise.

Was the ping successful? _____

Step 6: ping the Loopback IP address of this computer

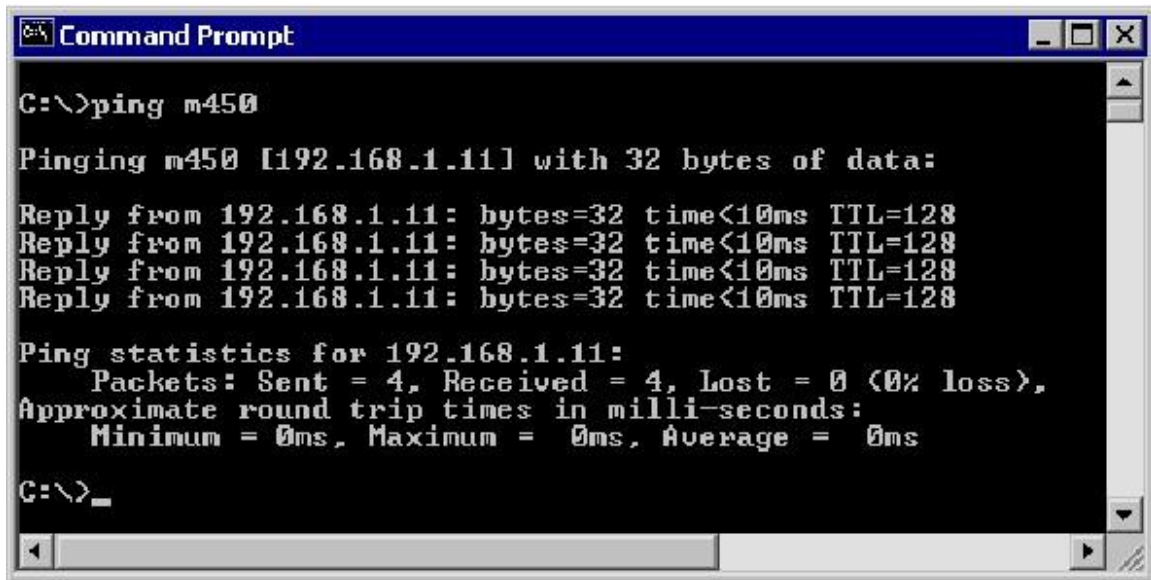
Type the following command: ping 127.0.0.1

The 127.0.0.0 network is reserved for loopback testing. If the ping is successful, then TCP/IP is properly installed and functioning on this computer.

Was the ping successful? _____

Step 7: ping the hostname of another computer

Try to ping the hostname of the computer that was recorded in the previous lab. The figure shows the successful result of the ping the hostname.



```
C:\>ping m450

Pinging m450 [192.168.1.11] with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<10ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>_
```

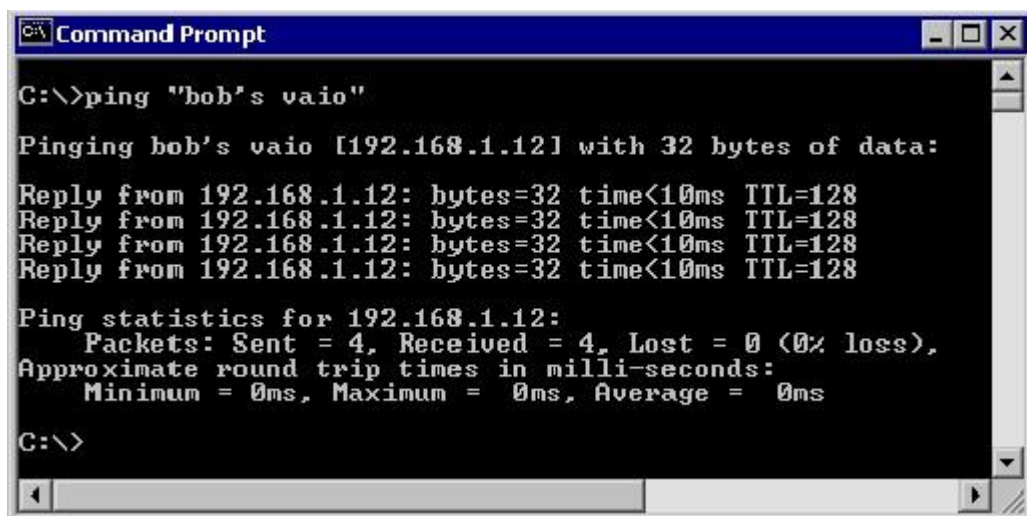
Fig 2.4 ping by host name on Command Prompt

Look over the results. Notice that the first line of output shows the host name, m450 in the example, followed by the IP address. This means the computer was able to resolve the host name to an IP address. Without name resolution, the ping would have failed because TCP/IP only understands valid IP addresses, not names.

If the ping was successful, it means that connectivity and discovery of IP addresses can be done with only a hostname.

If the last ping worked, try to ping the hostname of any another computer on the local network. The following figure shows the possible results.

Note: The name had to be typed in quotes because the command language did not like the space in the name.



```
C:\>ping "bob's vaio"

Pinging bob's vaio [192.168.1.12] with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<10ms TTL=128
Reply from 192.168.1.12: bytes=32 time<10ms TTL=128
Reply from 192.168.1.12: bytes=32 time<10ms TTL=128
Reply from 192.168.1.12: bytes=32 time<10ms TTL=128

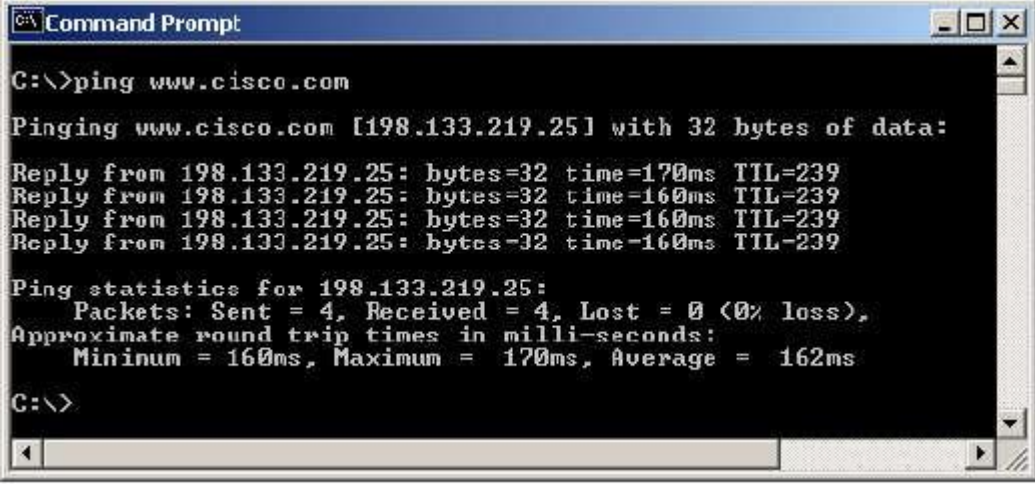
Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig 2.5 ping by host name with spacing on Command Prompt

Step 8: ping the Cisco web site

Type the following command: ping www.cisco.com



```
Command Prompt

C:\>ping www.cisco.com

Pinging www.cisco.com [198.133.219.25] with 32 bytes of data:

Reply from 198.133.219.25: bytes=32 time=170ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239
Reply from 198.133.219.25: bytes=32 time=160ms TTL=239

Ping statistics for 198.133.219.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 160ms, Maximum = 170ms, Average = 162ms

C:\>
```

Fig 2.6 ping by web address on Command Prompt

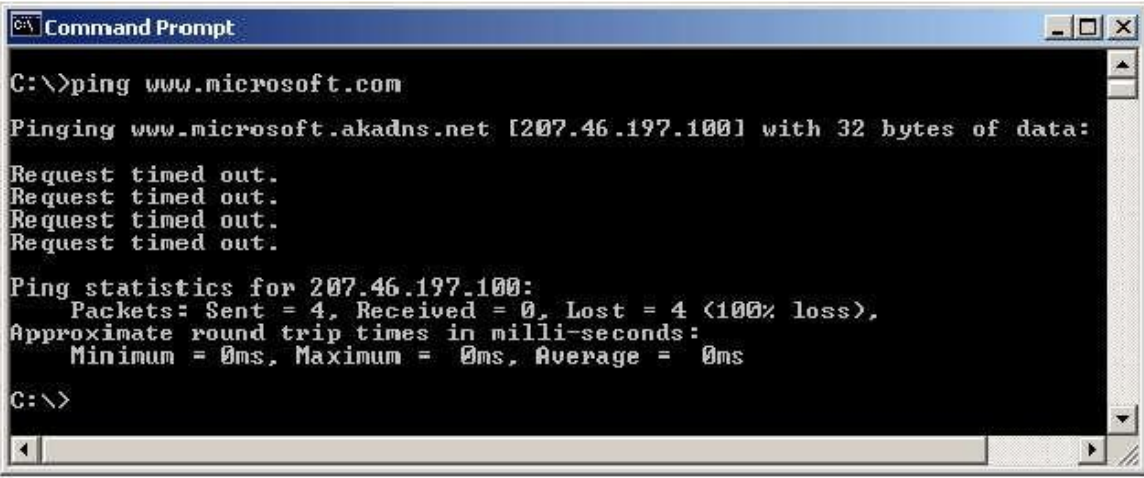
The first output line shows the Fully Qualified Domain Name (FQDN) followed by the IP address. A Domain Name Service (DNS) server somewhere in the network was able to resolve the name to an IP address. DNS servers resolve domain names, not hostnames, to IP addresses.

Without this name resolution, the ping would have failed because TCP/IP only understands valid IP addresses. It would not be possible to use the web browser without this name resolution.

With DNS, connectivity to computers on the Internet can be verified using a familiar web address, or domain name, without having to know the actual IP address. If the nearest DNS server does not know the IP address, the server asks a DNS server higher in the Internet structure.

Step 9: ping the Microsoft website

a. Type the following command: ping www.microsoft.com



```
Command Prompt

C:\>ping www.microsoft.com

Pinging www.microsoft.akadns.net [207.46.197.100] with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 207.46.197.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

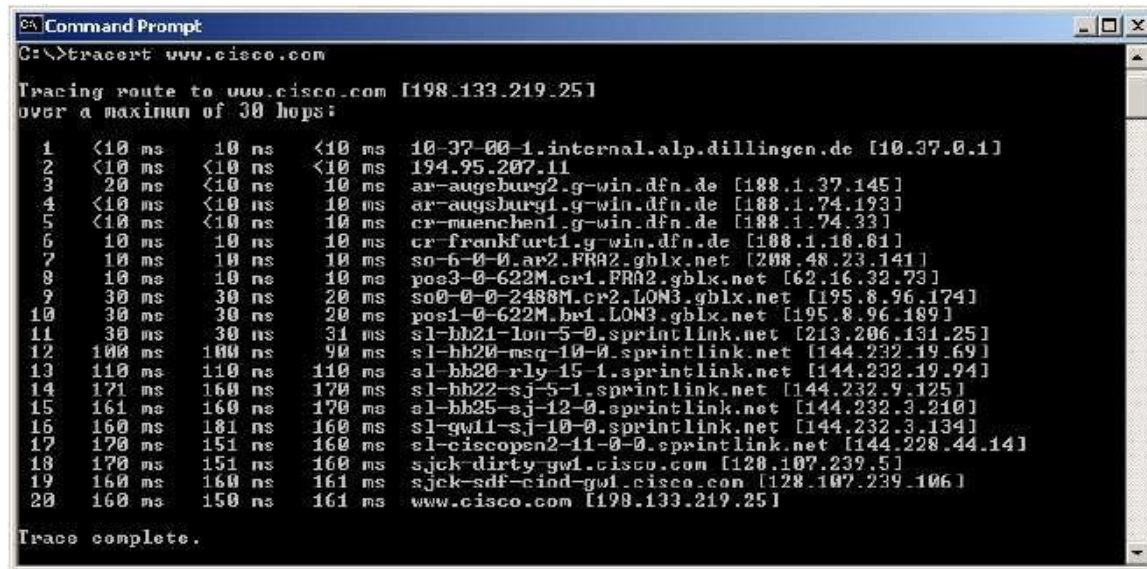
Fig 2.7 ping failure for Microsoft because some Microsoft routers are configured to ignore ping

Notice that the DNS server was able to resolve the name to an IP address, but there is no response. Some Microsoft routers are configured to ignore ping requests. This is a frequently implemented security measure.

ping some other domain names and record the results. For example, ping www.msn.de

Step 10: Trace the route to the Cisco web site

Type `tracert www.cisco.com` and press Enter.



```
Command Prompt
C:\>tracert www.cisco.com

Tracing route to www.cisco.com [198.133.219.251]
over a maximum of 30 hops:

  0  <10 ms  <10 ms  <10 ms  10.37.00.1.internal.alp.dillingen.de [10.37.0.1]
  1  <10 ms  <10 ms  <10 ms  194.95.207.11
  2  20 ms  <10 ms  <10 ms  ar-augsburg2.g-win.dfn.de [188.1.37.145]
  3  <10 ms  <10 ms  <10 ms  ar-augsburg1.g-win.dfn.de [188.1.74.193]
  4  <10 ms  <10 ms  <10 ms  cr-muenchen1.g-win.dfn.de [188.1.74.33]
  5  <10 ms  <10 ms  <10 ms  cr-frankfurt1.g-win.dfn.de [188.1.10.61]
  6  10 ms  10 ms  10 ms  so-6-0-0.ar2.FRA2.gblx.net [208.48.23.141]
  7  10 ms  10 ms  10 ms  pos3-0-622M.cr1.FRA2.gblx.net [62.16.32.73]
  8  30 ms  30 ms  30 ms  so0-0-0-2488M.cr2.LON3.gblx.net [195.8.96.174]
  9  30 ms  30 ms  30 ms  pos1-0-622M.br1.LON3.gblx.net [195.8.96.189]
 10  30 ms  30 ms  30 ms  sl-bb21-lon-5-0.sprintlink.net [213.206.131.251]
 11  100 ms 100 ms 100 ms  sl-bb20-msq-10-0.sprintlink.net [144.232.19.69]
 12  110 ms 110 ms 110 ms  sl-bb20-rlj-15-1.sprintlink.net [144.232.19.94]
 13  171 ms 160 ms 170 ms  sl-bb22-sj-5-1.sprintlink.net [144.232.9.125]
 14  161 ms 160 ms 170 ms  sl-bb25-sj-12-0.sprintlink.net [144.232.3.210]
 15  160 ms 181 ms 160 ms  sl-gw11-sj-10-0.sprintlink.net [144.232.3.134]
 16  170 ms 151 ms 160 ms  sl-ciscopen2-11-0-0.sprintlink.net [144.228.44.141]
 17  170 ms 151 ms 160 ms  sjck-dirty-gw1.cisco.com [128.107.239.5]
 18  160 ms 160 ms 161 ms  sjck-sdf-cind-gw1.cisco.com [128.107.239.106]
 19  160 ms 150 ms 161 ms  www.cisco.com [198.133.219.251]

Trace complete.
```

Fig 2.8 tracert website on Command Prompt

tracert is TCP/IP abbreviation for trace route. The preceding figure shows the successful result when running tracert from Bavaria in Germany. The first output line shows the FQDN followed by the IP address. Therefore, a DNS server was able to resolve the name to an IP address. Then there are listings of all routers the tracert requests had to pass through to get to the destination.

tracert uses the same echo requests and replies as the ping command but in a slightly different way. Observe that tracert contacted each router three times. Compare the results to determine the consistency of the route. Notice in the above example that there were relatively long delays after router 11 and 13, possibly due to congestion. The main thing is that there seems to be relatively consistent connectivity.

Each router represents a point where one network connects to another network and the packet was forwarded through.

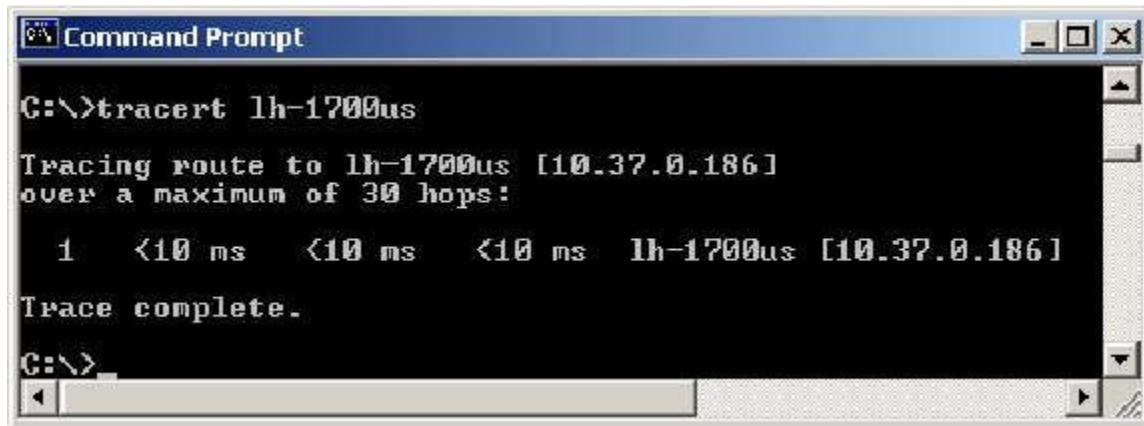
Step 11: Trace other IP addresses or domain names

Try tracert on other domain names or IP addresses and record the results. An example is tracert

www.hec.gov.pk

Step 12: Trace a local host name or IP address

Try using the tracert command with a local host name or IP address. It should not take long because the trace does not pass through any routers.



```
Command Prompt

C:\>tracert lh-1700us

Tracing route to lh-1700us [10.37.0.186]
over a maximum of 30 hops:

  1  <10 ms  <10 ms  <10 ms  lh-1700us [10.37.0.186]

Trace complete.

C:\>
```

Fig 2.9 tracert a nearby Router

TASKS:

Search and run the following commands:

- NSLOOKUP
- NETSTAT
- GETMAC
- HOSTNAME